Mukul Sharma

Curriculum Vitae · Machine Learning Scientist, perception, deep learning and foundation models

PERSONAL DETAILS

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GitHub	github.com/mukulsharma007
Work permit	Available (unrestricted)

PROFILE

Machine learning scientist (M.Sc. Computer Science, grade 1.9) with experience in computer vision, deep learning and foundation models. Built perception pipelines for engineering images and 3D CAD data, implemented neural networks from scratch in PyTorch, and fine-tuned large language models on HPC clusters. Experienced with Linux, Docker and SLURM, and with surveys for a user study. Strong interest in human-centered robotics and in robots that perceive and adapt to people.

EDUCATION

04/2024 – 06/2026 **M.Sc. Computer Science**
Leibniz University Hannover, Germany
Final grade: **1.9** (German scale, 1.0 = best)
Focus: machine learning, deep learning, natural language processing
Master's thesis: *Knowledge Graph-Filtered Reasoning-Augmented Generation for Factual Open-Domain Question Answering*

08/2015 – 10/2019 **B.Tech. Computer Science**
Dr. A.P.J. Abdul Kalam Technical University, India
Final grade: 78/100 (approx. German grade 2.3)
Bachelor's thesis: *University Recommendation System*

PROFESSIONAL EXPERIENCE

03/2025 – 06/2026 **Student Research Assistant**
Institute of Production Engineering and Machine Tools (IFW), Leibniz University Hannover

- Developed Python perception pipelines in a production engineering research project: YOLO-based object and symbol detection and OCR on technical drawings.
- Processed 3D geometry from CAD models (STEP files) to extract product information automatically.
- Prepared, structured and augmented image datasets (OpenCV); trained models on the HPC cluster (SLURM, NVIDIA GPUs) and improved detection performance and inference quality.

01/2020 – 03/2024 **Senior Analyst (Software)**
HCL Technologies, India

- Developed Python and JavaScript software across the full development lifecycle, with testing, debugging, code reviews and documentation.
- Built automated ETL and ELT data pipelines to ingest, clean, transform and preprocess data.
- Set up CI/CD with Git and Jenkins, containerised applications with Docker and deployed them on Google Cloud Platform.
- Developed machine learning models for predictive analysis and trend forecasting.

RESEARCH PROJECTS

M.Sc. research **Deep learning models implemented from scratch**

- Implemented linear and logistic regression, CNNs, RNNs, LSTMs and GANs in PyTorch from first principles, with systematic hyperparameter tuning and evaluation.

M.Sc. research **Fine-tuning and compressing foundation models**

- Fine-tuned open-weight language models (Mistral, Qwen) on NVIDIA GPUs with SLURM; reduced model size and inference cost with quantization and knowledge distillation.

Master's thesis	**Retrieval and evaluation for reliable question answering** • Built a system that retrieves evidence through semantic search and a knowledge graph and filters it with an LLM before answering; evaluated answer quality and faithfulness with RAGAS.
User study	**User study on IT authentication** • Conducted surveys with participants for a user study on IT authentication.

SKILLS

Perception	YOLO object detection, image segmentation, OCR, OpenCV, data augmentation; 3D CAD geometry (STEP)
Deep learning	PyTorch; CNN, RNN, LSTM and GAN models written from scratch; hyperparameter tuning
Foundation models	Large language models (Mistral, Qwen), fine-tuning, quantization, knowledge distillation, Hugging Face Transformers, retrieval-augmented generation
Programming	Python (primary), C, Java, JavaScript
Linux, Docker, HPC	Linux, Docker, HPC clusters with SLURM, NVIDIA GPUs, Git, CI/CD, Google Cloud Platform
User research	Surveys for user studies
Research	Literature reviews, reproducing published results, LaTeX/Overleaf

LANGUAGES

English: fluent (C1) · German: basic (A2)

REFERENCES

Referee 1	**Sandipan Sikdar** Junior Professor, L3S Research Center (Institute for Data Science), Leibniz University Hannover sandipan.sikdar@l3s.de · Academic reference
Referee 2	**Marcus Nein** Research Associate, Institute of Production Engineering and Machine Tools (IFW), Leibniz University Hannover Nein@ifw.uni-hannover.de · Reference for my student research assistant work

Hannover, 12 October 2026

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